



Molecular Adaptations to Concurrent Strength and Endurance Training

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Introduction

Both strength and endurance training regimens induce distinct skeletal muscle adaptations [1, 2]. It has been suggested that when performed in conjunction with a high frequency, volume, and intensity, conflicting adaptations may occur. If not implemented properly, concurrent strength and endurance training (CT) may result in reduced strength training (ST) adaptations [3–5]. The negative interaction between ST and endurance training (ET) has been defined as the interference effect. The advancements in technology have increased our understanding of the molecular mechanisms behind the exercise-induced adaptations to both strength and endurance stimuli. The molecular basis of muscular adaptation in response to different training regimens is very complex. It involves increases in the expression and/or activity of genes and proteins in order to increase contractile tissue (i.e., muscle hypertrophy) or mitochondrial content [6, 7]. Furthermore, there is considerable cross-talking and redundancy between signaling pathways that control exercise-induced adaptations, which is beyond the scope of this chapter (for further readings, see [6, 8, 9]). The following sections will review the acute and chronic molecular responses induced by CT, and how they are related to the interference effect and morphological changes.

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Molecular Interference Hypothesis

ST-induced muscle hypertrophy is the cumulative effect of multiple training sessions that have resulted in a net positive balance between the rate of muscle protein synthesis (MPS) and muscle protein breakdown (MPB) [10]. Despite the complexity of the protein synthesis process, it is regulated through signaling pathways that enhance mRNA translation [11]. The mammalian or mechanistic target of rapamycin complex 1 (mTORC1) is a key kinase controlling protein synthesis and muscle hypertrophy [12, 13]. The role of mTOR controlling other kinases involved in protein synthesis is explained in Chap. 5.

On the other hand, ET-induced adaptations are attributed to increases in mitochondrial function and content within the skeletal muscles that ultimately results in improved oxidative capacity and endurance performance [14–16]. Even though ET has been shown to activate multiple signals (Fig. 8.1), [17]. The peroxisome-proliferator-activated receptor γ co-activator-1 α (PGC-1 α) has been referred to as a key driver of mitochondrial biogenesis [18]. PGC-1 α is a transcriptional co-activator that induces mitochondrial biogenesis. It activates different transcription factors that modulate gene expression resulting in the encoding of mitochondrial proteins [19, 20]. Even though multiple signals can modulate PGC-1 α , it has been suggested that AMP-activated protein kinase (AMPK) is one of the major contributors as it controls the transcription and activity of PGC-1 α [21–23].

AMPK also plays an important role in cellular energy metabolism during exercise and nutrient deprivation [24]. Furthermore, AMPK has been referred to as a sensor for cellular energy status as it is activated by increases in cellular AMP:ATP ratio. Once activated, AMPK blunts the biosynthetic process that demands ATP consumption and stimulates an energy generating process in order to reestablish cellular energy levels [25]. In fact, animal model studies have demonstrated that AMPK activation can inhibit mTORC1 activity, its downstreams and blunts MPS and skeletal muscle hypertrophy [25–27].

In this regard, the greater energy demands and the different signaling pathways involved with CT compared to a single mode regimen create a conflicting environment within the skeletal muscle. Therefore, a molecular hypothesis has been put forward to explain the reduced training adaptation after CT [26]. In this hypothetical model referred to AMPK-Akt switch hypothesis, the AMPK phosphorylates tuberous sclerosis complex-2 (TSC2). TSC2 switches off the mTORC1-signaling cascade, ultimately decreasing the potential for muscle fiber hypertrophy after CT regimen [26] (Fig. 8.1).

Acute Molecular Responses of Concurrent Exercise

When analyzing the current literature in molecular responses to CT, the reader needs to consider multiple factors, as there is an important disparity in the experimental study designs (Table 8.1). For example, demographics, intra-individual responses, training status, exercise mode and volume, and nutrient availability will

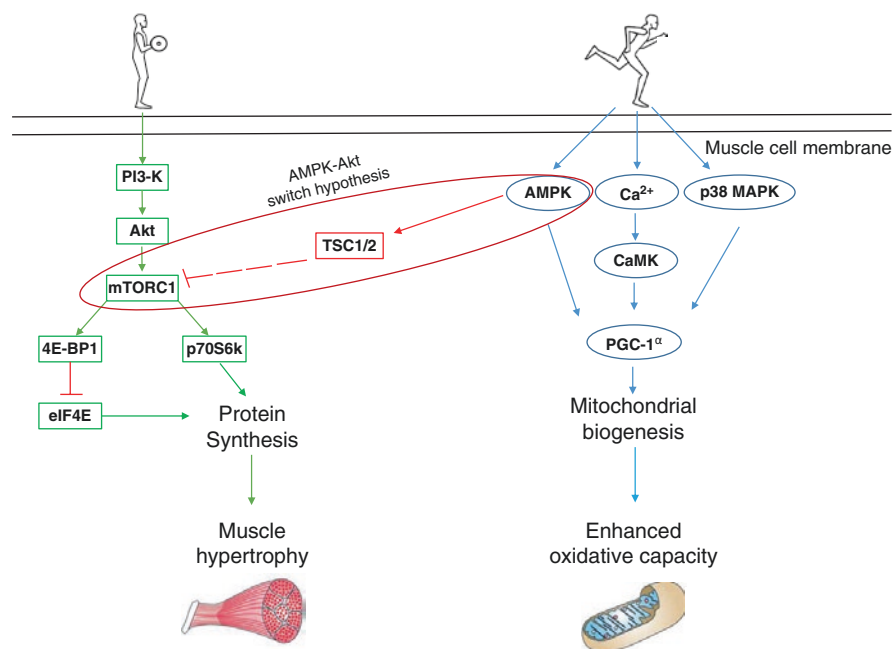


Fig. 8.1 Schematic diagram summarizing signaling pathways activated by strength (ST) and endurance training (ET) and the AMPK-Akt switch hypothesis. ST induces an increase in the activity of protein kinase B (Akt), mammalian target of rapamycin complex 1 (mTORC1), that modulates rates of protein synthesis through phosphorylation of eukaryotic initiation factor 4E-binding protein (4E-BP1) that promotes dissociation between 4E-BP1 and eukaryotic initiation factor 4E- (eIF4E) and activation of 70 kDa ribosomal protein kinase (p70S6K). ET activates signaling cascade that regulates metabolic process and mitochondrial biogenesis that comprises adenosine-monophosphate-activated protein kinase (AMPK), p38 mitogen-activated protein kinase (MAPK), calmodulin-dependent protein kinase (CaMK) and proliferator-activated receptor-gamma coactivator 1- α (PGC-1 α). In the Akt-AMPK switch hypothesis model, AMPK activated by ET may inhibit mTORC1 signaling cascade through tuberous sclerosis complex (TSC) blunting ST-induced protein synthesis. Figure was adapted from: Hawley, JA. (2009) Molecular adaptations to strength and endurance: Are they incompatible? Atherton et al. (2005). Selective activation of AMPK-PGC-1 or PKB-TSC2-mTOR signaling can explain specific adaptive responses to endurance or resistance training-like electrical muscle stimulation. Coffey and Hawley (2017). Concurrent exercise training: do opposites distract?

all affect the acute molecular responses. Furthermore, the muscle sampling time can significantly impact how the selected molecular markers will respond at the transcriptional level (i.e., gene expression) and the translational level (i.e., protein signaling and myofibrillar and sarcoplasmic synthetic rates).

Coffey and Hawley [7] have suggested that training status and intra-individual responses are the primary factors modulating the molecular cascades induced by CT [7]. These researchers investigated the effect of training background on acute molecular responses to strength exercise (SE) and endurance exercise (EE) in athletes with different training experience (i.e., either strength or endurance). In one experimental

Table 8.1 Summary of studies designs that investigate the acute molecular responses to concurrent exercise

Study	Training status	Study design	Nutrient availability	Sampling time	Exercise regimens				Conclusions
					SE	EE	Recovery between exercise	CT order	
Lundberg et al. [36]	Recreationally trained EE, 2–3×/week or SE 1–2×/week for more than a year	Intra-subject, unilateral cross-over, one leg performed SE, and the contralateral the CE	EE was performed 1 h after breakfast (1.01 g CHO/kg BM, 0.31 g protein/kg BM, and 0.24 g fat/kg BM), SE was performed 3 h after lunch (2.02 g CHO/kg BM, 0.62 g protein/kg BM, and 0.49 g fat/kg BM)	Baseline, 15 min and 3 h after SE	Unilateral leg-press and leg-extension, 2 × 7 each at maximal effort	~42 min, 40 min unilateral cycling ergometer at 70% of W_{max} , then workload was increased by 20 W until exhaustion	6-h	EE→SE	Performing SE 6 h after EE did not compromise signaling of mTOR-related proteins
Babcock et al. [32]	Recreationally active, college-aged males. EE 2×/week and SE 2×/week for >3 months	Intra-subject, unilateral cross-over, one leg performed SE, and the contralateral the CE	Experimental sessions were performed 2 h after subjects received standardized breakfast (500 kcal: 90–100 g of carbohydrates, 8–12 g of protein, and 4–8 g of fat)	Baseline, 96 h after exercise regimens	4 × 10 unilateral leg-extension at 75% of 1RM followed by unilateral leg-press 4 × 10 at the same intensity 75%.	90 min of cycle ergometer exercise at 60% of W_{max}	10 min	SE→EE	CE attenuates the satellite cell response to SE alone, and does so by preferentially blunting increases in MHC I satellite cell content

Coffey et al. [30]	Regular CT (>3x/week)	Randomized cross-over, all the participants performed two CT experimental sessions	Experimental sessions were performed after 10-h overnight fasting	Baseline, 0, 15 min and 3 h after sessions	8 × 5 at 80%, leg-extension exercise	10 × 6 s, all-out sprints, cycling ergometer against 0.75 Nm torque ⁻¹ kg ⁻¹	15 min	Both orders were performed	Repeated sprints impair anabolic response induced by SE and enhance catabolic responses
Coffey et al. [31]		Randomized cross-over, all participants performed two CT experimental sessions	Experimental sessions were performed after 10-h overnight fasting	Baseline, 0, 15 min and 3 h after sessions	8 × 5 at 80% of 1RM, leg-extension exercise	30 min at 70% of VO _{2peak} , cycling ergometer	15 min	Both orders were performed	CE produce suboptimal signaling responses associated with single mode exercises
Wang et al. [33]	Recreationally active	Randomized cross-over, all participants completed the two experimental sessions	Experimental sessions were performed after 12-h overnight fast	Baseline, 1 and 3 h after exercise	6 sets at 70, 75, 80, 80, 75, 70% of 1RM. Participants were oriented to complete maximal reps as possible up to 15 on the leg-press exercise	60 min continuous cycling at 65% VO _{2max} (3-min rest allowed after 30 min)	15 min	EE→SE	Performing SE after EE augmented signaling of mitochondrial biogenesis
Pugh et al. [35]	Healthy untrained males	Counterbalanced cross-over design	2 h prior the experimental sessions, a breakfast was provided (1803 kJ, 55% carbohydrate; 29% fat and 16% protein)		4 × 8 at 70% of 1RM on leg-extension exercise	10 × 1 min at 90% HR _{max}	2 min after the last SE set	SE→EE	CE utilizing low volume and high intensity EE does not attenuate anabolic protein signaling and mRNA expression, or exacerbates marker of protein breakdown in fed state

(continued)

Table 8.1 (continued)

Study	Training status	Study design	Nutrient availability	Sampling time	Exercise regimens			Recovery between exercise	CT order	Conclusions
					SE		EE			
Donges et al. [39]	Sedentary middle-aged men	Randomized cross-over, all participants completed three experimental sessions, SE, however, CE comprised 50% volume of SE and EE alone	Sessions were performed after 10-h overnight fast and subjects received 20 g of whey protein immediately after the experimental session	Baseline, 1 and 4 h after exercise	8 × 8 at 70% of 1RM on the leg-extension exercise or 4 × 8 for CE condition	40 min of continuous cycle ergometer at 55% of $\dot{V}O_{2peak}$ or 20-min for CE		None	SE→EE	When providing 20 g of whey protein immediately after the training sessions, CE performed with 50% less training volume is effective as SE stimulating both myofibrillar and mitochondrial protein synthesis fractions
Aprio et al. [43]	Moderately trained male	Randomized cross-over, all participants completed three experimental sessions	Sessions were performed after ~10.5 h overnight fast	Baseline, 1 h and 3 h after	10 × 10 on leg-press exercise (4 × 10 at 85% of 1RM, 4 × 12 at 75% of 1RM, and two sets to failure at 65% of 1RM	30 min at 70% of $\dot{V}O_{2max}$		15 min	EE→SE	EE-induced elevation in AMPK does not blunt anabolic signaling pathway after subsequent SE

Table 8.1 (continued)

Carrithers et al. [38]	Moderately active	Intra-subject, unilateral cross-over, one leg performed SE, and the contralateral the CE	Sessions were performed after 10.5–14.5 h overnight fast	Baseline, immediately and 3 h after	3 × 10 at 80% of 1RM for each exercise, unilateral leg-press and leg extension, and a fourth set to failure	90 min at 60% of their peak power while maintaining a cadence of 60 rpm.	30 min	EE→SE	While it was reported that CE does not suppress post-SE myofibrillar protein synthesis. There were no differences across time and conditions, which may suggest that exercise stimulus was not sufficient to alter the protein turnover
Lambert et al. [37]	Overweight/obese individuals	Parallel group repeated measures design with acute and chronic assessments, where each group performed one of three acute sessions: SE, CE, and CE aquatic and CE aquatic which EE was performed in aquatic treadmill	Sessions were performed after 12 h overnight fast	Baseline and 24 h after exercise	3 × 12 at 60% of 1RM for each exercise, leg-press, leg-curl, and leg-extension	Time to complete 250 kcal at 60% of VO_{2max}	None	SE→EE	CE utilizing aquatic treadmill elicited greater myofibrillar protein synthesis responses compared to SE and conventional CE (i.e., EE performed on land treadmill)

SE strength exercise, EE endurance exercise, CE concurrent exercise, BM body mass, CHO carbohydrate, W Watts, W_{max} peak power output, 1RM one repetition maximum, VO_{2peak} peak oxygen uptake, VO_{2max} maximum oxygen uptake, HR_{max} maximum heart rate, KJ kilojoules

session, the athletes underwent their typical exercise regimen. In the following session, they switched, undertaking the unfamiliar exercise regimen. The authors found that AMPK^(Thr172) phosphorylation increased after EE in strength-trained but not in endurance-trained individuals. On the other hand, AMPK^(Thr172) and p70S6K^(Thr389) increased after SE in the endurance athletes but not in strength-trained individuals [28]. It is important to point out that AMPK has been elevated after SE in untrained individuals [29]. This suggests that AMPK may not be exclusively activated by EE. Therefore, the main takeaway is that training background diminishes sensitivity to specific molecular responses associated with single mode exercise.

Regarding acute molecular responses to concurrent strength and endurance exercises (CE), conflicting results have been observed. While previous research has demonstrated that CE can decrease the anabolic signaling pathways and myogenic factors when compared to SE [30–32], other studies have demonstrated that CE produced similar and/or favorable responses on endurance [33] and anabolic signaling pathways [34–36] and on MPS [37–39].

For example, Lundberg et al. [36] found favorable results on selected gene expression and protein signaling after CE compared to SE in recreationally active males. They reported that PGC-1 α and VEGF (i.e. an angiogenic factor) gene expression were higher in CE compared to SE. Regarding gene expression of MuRF-1 and Atrogin-1 that are considered to be the major modulators of MPB. While, MuRF-1 demonstrated a similar decrease across conditions. Atrogin-1 decreased over time only in CE. Myostatin, which inhibits protein synthesis demonstrated lower mRNA levels in CE. The authors also reported favorable results in mTOR-p70S6K axis. There was a significant exercise effect, in which mTOR^(Ser2448) and p70S6K^(Thr389) phosphorylation were higher in CE. These findings are well in line with previous studies and suggest that a bout of CE can increase anabolic signaling pathways in an acute setting [34] with higher or no suppression on MPS [37–39].

Nevertheless, some researchers have pointed out an important and inherent limitation when comparing the results of studies that have investigated single mode exercise versus CE [7]. Those authors stated that CE often has imposed a greater contractile stimulus to the skeletal muscle as during a CE session typically more work is performed by combining the two single mode exercises [7]. This additional work may stimulate protein signaling and MPS to a greater extent compared to single mode exercises. In order to address this issue, Donges and colleagues [39] had sedentary middle-aged men to perform single modes of SE and EE, as well as a CE session, which had 50% less volume than each single mode exercise session (i.e., all sessions were matched for volume). The authors found differences in protein signaling that may indicate favorable responses to SE. They also reported that mitochondrial protein synthetic rate increased in a similar fashion between the three exercise regimens. In addition, MPS was similar for both SE and CE and they were both greater than in EE alone [39]. However, one important factor to consider is that the participants consumed 20 g of whey protein isolate immediately after exercise. This was done in order to maximize the anabolic potential in response to exercise regimens. However, this makes it impossible to verify if CE and SE sessions modulated protein synthetic responses in a different fashion.

The above-mentioned contention is supported by different evidences: (1) studies have demonstrated that 20 g of protein is sufficient to maximize MPS after SE [40, 41], and (2) it has been demonstrated that whey protein ingestion after CE produced greater MPS and attenuated markers of MPB compared to CE + placebo [42]. In addition, other studies that have investigated the effects of CE on protein synthetic responses did not provide conclusive findings. For example, when EE comprised aquatic treadmill and the work performed was the double than SE alone (i.e., 500 vs. 250kcal, respectively), MPS favored to CE in overweight/obese individuals [37]. Another study suggested that MPS after CE was not suppressed when compared to SE, as both regimens demonstrated similar rates of protein synthesis. Nonetheless, neither exercise regimen demonstrated significant increases in MPS compared to baseline [38], further suggesting that the exercise regimens did not produce enough strain to stimulate anabolic responses. Therefore, more research needs to be done assessing the effects of CE and SE on protein synthetic responses before any general conclusions can be drawn.

On the other hand, some studies have suggested that CE may attenuate acute anabolic signaling and myogenic responses [30–32]. While this data may suggest some acute interference, it is important to point out that studies demonstrating that are limited [30–32]. Coffey et al. [30] submitted recreationally trained individuals to perform two different orders of CE (i.e., SE→EE and EE→SE). In general, the authors reported modest changes in protein signaling phosphorylation regardless of the exercise order. However, p70S6K^(Thr389) was elevated to the greatest extent when the strength bout was performed before sprints bouts. The authors also investigated the selected genes associated with myogenesis and proteolysis activity. The gene expression of insulin-like growth factor I (IGF-I) and MyoD did not demonstrate any order effects other than an inhibited gene expression of IGF-1. In regard to the marker of proteolysis, Atrogin-1 demonstrated no significant differences, whereas MuRF-1 gene expression was elevated 3-h post the CE regardless of the exercise order [30]. In another study, the authors also found results that suggested some acute interference. They observed a significant reduction in the mTOR^{Ser2448} phosphorylation levels when EE was performed after SE, but not in the opposite order [31].

These findings from Coffey and colleagues [30, 31] suggest that EE hampered anabolic signaling pathways in response to SE. Furthermore, it seems that CE exacerbates proteolytic gene expression in the acute fashion in recreational trained individuals. These two studies performed by Coffey et al. [30, 31] demonstrate strong ecological validity (i.e., real-world generalizability) as the study design represents well how different athletes and trained individuals perform CE. However, these studies also have inherent limitations: The lack of a positive control condition (i.e., strength exercise alone) makes it difficult to conclude whether CE produced interference on the translational signaling pathways and proteolytic markers. Moreover, analyzing selected mRNA expression may not reflect the protein levels in response to CE. Although nutrient availability might be a potential candidate for the outcomes reported by the authors, there is conflicting evidence from other studies [43, 44]. These studies involved trained participants performing CE after 9–10 h of fasting, and they did not report negative effects on anabolic signaling pathways [43, 44].

Other potential mediators in the molecular adaptations to exercise are satellite cells. Satellite cells function as donors of new myonuclei for the muscle fiber and are considered to be myogenic precursors. They play an important role in muscle regeneration, repair and possibly also aid in muscle hypertrophy as a response to ST [45]. In fact, one study has demonstrated that those individuals considered high responders for muscle fiber hypertrophy demonstrated a greater satellite cell number after the training period [46].

To date, only one study has investigated the acute effects of CE on satellite cells [32]. Babcock et al. [32] submitted recreationally active individuals to different exercise regimens. While SE increased satellite cell density more than CE in type II muscle fibers. It is important to mention that the elevated baseline satellite cell count for CE may have decreased the potential for additional increases following CE. However, satellite cells increased by 46% after SE, 7% after EE, and 8% after CE in muscle fiber type I. These results are interesting, as it has been consistently demonstrated that the interference effect at a morphological level (i.e., muscle hypertrophy) occurred primarily among muscle fiber type I [3, 4, 47]. In addition, the findings by Babcock et al. [32] suggested that the addition of ET suppressed the increase of satellite cells induced by SE. Again, this should be interpreted cautiously as EE alone might activate satellite cells for muscle remodeling as well [48, 49]. Thus, more studies are necessary in order to elucidate the effects of CE on satellite cells activation.

In light of incongruences in study designs, the anabolic molecular responses seem to be very similar between SE and CE. Furthermore, AMPK downregulating mTORC1 has been consistently demonstrated in animal models [25, 26]. However, the proposed hypothesis for the interference effect at the molecular level in which AMPK switches off mTORC1 has not been confirmed in human studies. In fact, studies have demonstrated that increases in AMPK phosphorylation induced by EE did not blunt mTORC1-signaling pathway or MPS responses [43]. In addition, satellite cells and markers of MPB emerge as potential factors explaining the acute molecular interference in CE. However, only one study has investigated the effects of SE and CE on satellite cells population. Also, scrutinizing mRNA expression may not be representative of protein content and it is difficult to differentiate whether those changes are associated with MPB or greater protein turnover. Although these acute studies have shed some light on the variation of molecular responses to CE, it remains unclear on how this may translate to chronic molecular and morphological adaptations induced by CE.

Chronic Molecular Adaptations to Concurrent Training

Chronic adaptations to ST and ET are likely a result of cumulative changes in gene expression, acute signaling, and protein synthesis/breakdown. This ultimately leads to changes in the phenotype, morphology, and functionality of skeletal muscle [5, 6, 50–52]. Even though skeletal muscle displays a remarkable ability to adapt to different exercise stimuli, it has been demonstrated that CT may negatively affect the adaptations induced by ST in isolation (i.e., muscle hypertrophy, strength, and power) [3,

4, 53–56]. From a molecular standpoint, it has been suggested that the conflicting adaptations are a result of distinct signaling responses induced by ST and ET [26, 57, 58] (Fig. 8.1). As previously mentioned, the signaling cascade activated by ET can inhibit the anabolic stimulus of ST compromising long-term adaptations induced by ST. While the number of studies investigating the acute responses to concurrent exercise are low, there are even fewer studies available on the chronic effects of CT [34, 36, 37, 56, 59–61]. Additionally, there are significant differences in the chronic experimental designs, which should be taken into account when attempting to make comparisons between investigations (Table 8.2). Nevertheless, this section will address the methodological differences between these studies and extrapolate the chronic effects of CT on molecular adaptations and morphological changes based off of a comprehensive analysis of the available literature on the topic.

Regarding the molecular makers associated with ET adaptations, one study found no significant changes on PGC-1 α and VEGF mRNA expression after 5 weeks of CT and ST [62]. On the other hand, another study from the same researchers reported that after 5 weeks of training, CT induced a significant reduction on PGC-1 α mRNA expression. In addition, VEGF mRNA expression was higher in CT than ST [34]. There may be some methodological discrepancies that could explain the different findings in these investigations. First off, mRNA is relatively short-lived and unstable, and Lundberg et al. [62] performed the muscle biopsies 72 h after the last training session that may have diminished the ability to detect significant changes on those selected genes [63]. Conversely, Fernandez-Gonzalo et al. [34] performed muscle biopsies 3 h post ST regimen, nevertheless there was a significant group effect (i.e., CT greater than ST) probably because of the higher levels of PGC-1 α at baseline for CT regimen.

In addition, PGC-1 α mRNA increased in response to ET, ST, and CT regimens, while PGC-1 α 4 increased only after ST and CT. Moreover, CT demonstrated the greatest increase in PGC-1 α 4 compared to all the other groups [60]. These results suggest that other PGC-1 α isoforms might be associated with the long-term adaptations induced by CT. Different from PGC-1 α that is associated with mitochondrial changes inducing oxidative capacity improvements, PGC-1 α 4 is associated with increases in muscle size [60, 64]. Therefore, more research is necessary in order to elucidate the role of different PGC-1 α isoforms on exercise-induced adaptation.

Another issue that limits our understating on molecular adaptations to CT is that only a few studies have compared the long-term effects of CT to ET in isolation on protein signaling responses. We [59] investigated the effects of ST, ET, and CT on basal AMPK^(Thr172) phosphorylation [59]. After 8 weeks of training, the intermittent ET regimen was the only group to demonstrate significant increases in AMPK^(Thr172) when compared to baseline levels. On the other hand, Wilkinson et al. [52] reported that after 10 weeks of either continuous ET or ST in isolation, both training regimens increased AMPK^(Thr172). However, AMPK^(Thr172) returned to baseline levels 4 h later on both regimens. Importantly, ET was the only group that significantly increased mitochondrial protein synthetic rates above resting values. In agreement with the acute data, those results suggest that AMPK^(Thr172) might not be specific to exercise modalities [65].

Table 8.2 Summary of study designs that investigate the chronic molecular and morphological adaptations to concurrent training

Study	Training status	Study design	Nutrient availability	Sampling time	Training regimens			Recovery between regimens	Conclusions
					ST	ET	CT		
Lundberg et al. [62]	Recreationally trained involved in team sports activities	Intra-subject, unilateral repeated measures design, one leg performed ST, and the contralateral the CT regimen. Five weeks training period	Biopsies were performed 2 h after subjects receiving a standardized breakfast, (1.01 g CHO/kg BM, 0.31 g protein/kg BM, and 0.24 g fat/kg BM	Pre-training, 72 h after the last training session	2 days/week—12 sessions: 4 × 7 at unilateral knee extensions in the flywheel ergometer at maximal effort	Only as part of CT regimen: continuous ~42 min, 40 min unilateral cycling ergometer at 70% of W_{max} , then workload was increased by 20 W until exhaustion	3 days/week—15 sessions, ET followed by ST	6 h	5 weeks of CT and ST produced similar molecular adaptations on markers of ET adaptation and protein breakdown. Muscle volume increase was greater in CT regimen than ST. Muscle fiber CSA increase was not different between CT and ST.

Fernandez-Gonzalo et al. [34]	Recreationally trained involved in team sports activities	Intra-subject, unilateral repeated measures design, one leg performed ST, and the contralateral the CT regimen. Five weeks training period	EE was performed 1 h after breakfast (1.01 g CHO/kg BM, 0.31 g protein/kg BM and 0.24 g fat/kg BM), SE was performed 3 h after lunch (2.02 g CHO/kg BM, 0.62 g protein/kg BM and 0.49 g fat/kg BM)	Pre-training and 3 h after the last training sessions. Pre-test biopsies were performed prior ST, 6 h after completion of ET	2 days/week—12 sessions: 4 × 7 at unilateral knee extensions in the flywheel ergometer at maximal effort	Only as part of CT regimen: continuous ~42 min, 40 min unilateral cycling ergometer at 70% of W_{max} then workload was increased by 20 W until exhaustion	3 days/week—15 sessions, ET followed by ST	6 h	5-week training blunts the overall interference of greater anabolic response observed in acute fashion after CT in untrained muscle, which suggests that ET acts synergistically with ST only during short-term training.
Ruas et al. [60]	Sedentary	Parallel group repeated measures design. Participants were randomly assigned to C, ET, ST, or CT groups. Eight weeks training period	Not mentioned	Pre-training, 48 h after the last training session	Whole body routine, (week-1, light-weight, 4 days/week), (week-2, 2 × 8–10RM, 4 days/week), (week-3, 3 × 8–10RM, 4 days/week), (week-4–8, 4 × 8–10RM, 4 days/week)	Continuous stationary cycling at 65% of VO_{2max} , (week-1, 30 min, 3 days/week), (week-2, 45 min, 4 days/week), (week 3–8, 60 min, 5 days/week)	Progression of the ET was the same, except that the durations were half as long as the ET. The number of lifts was less for ST component in CT regimens as well	Not mentioned	8 weeks of CT augments mRNA gene expression associated with muscle size (e.g., PGC1 α) compared to ST and ET.

(continued)

Table 8.2 (continued)

Study	Training status	Study design	Nutrient availability	Sampling time	Training regimens				Recovery between regimens	Conclusions
Lambert et al. [37]	Overweight/obese individuals	Parallel group repeated measures design. Participants were randomly assigned to ST, ST + LTM, and ST + ATM groups. Twelve weeks training period	Sessions were performed after 12 h overnight fast	Baseline and 24 h after exercise	ST Whole body routine, 2 days/week. Week 1–6, 3 × 12 at 60% of 1RM. Week 8–12, 3 × 8 at 75% of 1RM.	ET Only as part of CT regimens. 3 days/week Continuous exercise Week 1–6, 250 kcal at 60–85 of VO_{2max} . Week 8–12, 500 kcal at 85% of VO_{2max} .	CT Progression of the ET and ST was the same. However, CT regimens performed three sessions of ET. ST was performed by ET regimens	None	12 weeks of CT regimen utilizing aquatic treadmill produced similar myofibrillar protein synthesis adaptations compared to traditional CT and ST. However, aquatic CT produced greater lean mass accrual.	

De Souza et al. [59] and De Souza et al. [56]	Recreationally active involved in team sports activities	Parallel group repeated measures design. Participants were randomly assigned to C, ET, ST, and CT groups. Eight weeks training period	Biopsies were performed 2 h after subjects receiving a standardized breakfast, (~311 kcal; 63.5%CHO, 21.8% protein, and 14.7% fat	Baseline and 48 h after exercise	Lower body routine (leg-press, knee extension and knee-flexion. 2 days/week. Week 1–2: 3 × 12RM, week 3–4: 4 × 8–10RM, week 5–6: 5 × 6–8RM, week 7–8: 3 × 10–12RM	Interval training, week 1–2: 20 × 1 min at 80 of vVO _{2max} , week 3–4: 20 × 1 min at 85–90% of vVO _{2max} , week 5–6: 20 × 1 min at 95% of vVO _{2max} , week 7–8: 20 × 1 min at 95–100% of vVO _{2max}	Progression of ST and ET was the same. The order in which the participants performed the strength and aerobic exercises within sessions was balanced.	No more than 5 min	While 8 weeks of training produced similar MSTN pathway and quadriceps CSA adaptations between CT and ST. CT produced interference in anabolic signaling and muscle fiber adaptations. ET was the only training to increase AMPK after the training period.
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Table 8.2 (continued)

Study	Training status	Study design	Nutrient availability	Sampling time	Training regimens			Recovery between regimens	Conclusions
Lundberg et al. [66]	Moderately trained involved in recreational team sports activities	Intra-subject, unilateral repeated measures design, one leg performed ST and the contralateral the CT regimen. Five weeks training period	Biopsies were performed 2 h after subjects receiving a liquid formula (1.01 g carbohydrates/kg bw, 0.31 g protein/kg bw, and 0.24 g fat/kg bw). The liquid formulas contained 6.3 g protein (0.55 g leucine), 20.2 g carbohydrates, and 4.9 g fat per 100 ml	Baseline, 3 h after the first training session, 72 h after last training session	ST 2 days/week—12 sessions: 4 × 7 extensions in the flywheel ergometer at maximal effort	ET Only as part of CT regimen: continuous ~42 min, 40 min unilateral cycling ergometer at 70% of W_{max} , then workload was increased by 20 W until exhaustion	CT 3 days/week—15 sessions. ET followed by ST	15 min	5 weeks of exercise inducing AMPK activation does not compromise muscle hypertrophic adaptation in CT compared to ST.

Fyfe et al. [61]	Recreationally involved in endurance and/or strength training at least twice a week for >30 min	Parallel group repeated measures design. Participants were randomly assigned to HIIT+ST, MICT+ST, or ST groups. Eight weeks training period	Biopsies were performed after ~8–10 h overnight fast	Baseline and 1 and 3 h after first and last training session	3 days/week—24 sessions: 3–4 × 4–14RM at ~65–90% of 1RM whole-body routine- 3 lower body exercise per training session (e.g., day 1 and 3—leg-press, leg-extension, leg curl exercise; day-2—leg-press, dumbbell lunges, and leg-curl exercise	Only as part of CT regimen: HIIT: 5–10 × 2 min intervals at 120–150 of lactate threshold; MICT: 15–33 min at 80–100% of lactate threshold	3 days/week. Two concurrent training regimes (HIIT and MICT). ET followed by ST	10 min	8 weeks of CT regimens favored adaptations on selected molecular markers indicating ribosome biogenesis and translational capacity. However, CT diminished chronic anabolic signaling and produced interference on muscle fiber CSA type I development.
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ST strength training, ET endurance training, CT concurrent training, C control, CHO carbohydrate, BM body mass, W Watts, W_{max} peak power output, VO_{2max} maximum oxygen body mass, CSA cross-sectional area, VO_{2max} maximum oxygen uptake, LTM land-treadmill, ATM aquatic-treadmill, MSTN myostatin, VO_{2max} velocityat maximal oxygen uptake, HIIT high-intensity interval training, MICT moderate-intensity continuous training

Nevertheless, the great debate is whether the AMPK-Akt switch hypothesis explains the chronic interference effect, and how CT and ST influence molecular parameters that control MPS and MPB. Currently, the chronic studies have demonstrated similar effects of ST and CT on AMPK phosphorylation levels [59, 61, 66]. Interestingly, different CT regimens produced an interference in ST-induced adaptations regardless of ET intensity employed (i.e., continuous or intermittent). The ST demonstrated the greatest AMPK activation compared with CT regimens after the last training session. Which suggests that rather than placing more energy stress on the muscle cell, chronic CT would allow muscles to be less susceptible to energy homeostasis perturbation induced by exercise [61]. While it is attractive to rule out that AMPK may help explain the interference effect, more evidence on how AMPK would respond during CT-induced interference effect are indeed warrant. To date, it is important to note that AMPK switching off mTORC1 has not been confirmed in chronic CT studies even when interference effect was reported [61].

In addition, the chronic studies, which have investigated the effects of CT on molecular and morphological adaptations have demonstrated that CT can enhance [34, 37, 62, 66], blunt [56], or produce a similar magnitude of muscle hypertrophy [59]. Importantly, as only few studies have investigated the chronic effect of CT on anabolic and MPB markers [34, 56, 60, 62], no clear pattern emerges in regard to the chronic molecular adaptations. Interestingly, while studies have suggested that CT can be more “catabolic” in an acute fashion [30], some chronic training studies have not confirmed this [34, 62]. For example, CT regimen was the only condition to demonstrate significant reductions in gene expression associated with protein breakdown, whereas ST regimen was the only one to demonstrate a significant reduction on Myostatin gene expression [34]. In fact, the effects of CT on Myostatin gene expression are scarce and controversial. For example, Lundberg and colleagues [62] did not report any significant changes on Myostatin levels. Similarly, our own study [56] also failed to report significant changes in Myostatin mRNA levels after 8 weeks of CT and ST. However, the SMAD-7 mRNA that negatively modulates the Myostatin gene expression significantly increased in both training regimens [56]. Ruas and colleagues [60], on the other hand, found that Myostatin mRNA expression was reduced only in ST and CT groups. Notwithstanding Myostatin is an important negative regulator of muscle mass, it is noteworthy to mention that Myostatin cannot differentiate muscle hypertrophy levels in humans [67]. Further studies are required in order to elucidate the long-term effects of CT and ST on proteolytic markers. The differences between protocols and the lack of data limit our understanding of the molecular markers that are associated with MPB in humans.

In regard to the chronic effects of CT on anabolic responses, the studies have demonstrated conflicting results as well [34, 37, 59]. Lambert et al. [37] is the only study that investigated the long-term effects of CT and ST on MPS. These researchers submitted overweight/obese individuals to three different training regimens. One CT group performed ST combined with ET on land treadmill (ST + LTM), while another CT group performed ST combined with ET on aquatic treadmill (ST + ATM), and the third group consisted of single mode ST. After a 12-week

training period, ST + ATM demonstrated the highest mean for myofibrillar protein synthetic rate, but no significant differences between groups. In addition, ST + ATM demonstrated the greatest lean body mass gains when compared to other two groups. Furthermore, Fernandez-Gonzalo et al. [34] did not report within or between groups on selected anabolic protein signaling (e.g., mTOR^{Ser2448}, rp6^{Ser235/236}, eEF2^{Thr56}), while p70S6^{Thr389} increased in similar fashion in CT and ST regimens. As part of this study was published elsewhere, the authors also reported that the CT leg demonstrated greater increases in muscle volume (i.e., CT: 13.6; ST: 7.8%) and muscle fiber cross-sectional area (i.e., CT: 17.0%; ST: 9.0%) [62]. Collectively, those studies provide evidence that CT might produce an additional effect on muscle hypertrophy with similar molecular adaptations between regimens.

Conversely, two other studies combined suggest that ST favors muscle hypertrophy and CT induced a down-regulation in the basal phosphorylation of Akt/mTORC1/p70S6K [56, 59]. De Souza et al. [59] reported that ST and CT increased p70S6K protein content. In addition, they found a significant group by time interaction which the ST group was the only group to increase basal Akt^(Ser473) and demonstrated a strong trend to towards significant p70S6K^(Thr389) phosphorylation. In the second study using the same sample, the authors found that ST was the only group that significantly increased cross-sectional area of type I and IIA, whereas the CT blunted muscle hypertrophy across the whole spectrum of fibers [56]. These two studies suggest that the interference effect might be associated with down-regulation of Akt/mTORC1/p70S6K signaling cascade.

These outcomes should be cautiously interpreted. In fact, some studies have demonstrated that ET can activate anabolic signaling pathways and MPS, specifically with cycling ergometer exercise in untrained individuals [68–71]. In addition, one review reported that several studies found significant increases in muscle size after ET utilizing a cycling ergometer modality in untrained individuals [64]. Interestingly, the studies that have demonstrated favorable chronic adaptations to CT were in untrained and/or physically active participants and it was used cycling ergometer modality [34, 62, 66]. Due to the subject's low training status, they are more sensitive to activate anabolic signaling pathways and demonstrated longer MPS responses [28]. Therefore, adding this ET stimulus with a greater resistance component to their short-term and low-volume ST scheme can make these CT regimens more anabolic than performing ST alone. Another important factor to be considered is the exercise mode, while it remains to be determined, evidence has suggested that ET performed with treadmill can be more detrimental to muscle hypertrophy during CT [72]. Even though Lambert et al. [37] used treadmill mode, they reported that utilizing the aquatic-treadmill exercise demonstrated greater acute MPS than the land treadmill. This culminated in greater lean body mass in the aquatic-treadmill group after the CT period. This is likely due to the increased resistance and reduced fatigue of the ATM modality. On the contrary, higher intensity interval training performed with the standard land treadmill demonstrated lower basal anabolic signaling which culminated in blunted muscle hypertrophy [56, 59].

However, an elegant study performed by Fyfe et al. [61] partially corroborates with the aforementioned findings. These authors were the first to investigate ribosome

biogenesis markers (i.e., gene expression and RNA content), and anabolic signaling in response to ST and CT regimens performed on a cycle ergometer in trained individuals. In a scenario which CT regimens produced attenuation in muscle fiber type I hypertrophy. The selected acute and chronic changes indicating ribosome biogenesis and translational capacity favored CT regimens regardless of the intensity (i.e., continuous and high-intensity). Nevertheless, the anabolic signaling in response to last training session was different. For example, mTOR phosphorylation was increased post-1 h only in ST and it was increased post-3 h only in the CT regimen which used high intensity ET. Still, changes in p70S6K^(Thr389) and rps6^{Ser235/236} were greater for ST when compared with both CT regimens. These findings suggest that (1) there is a disconnect between makers of ribosome biogenesis/translational efficiency and muscle fiber hypertrophy, (2) the chronic repetition of the training stimulus may allow skeletal muscles to be more mode-specific, and (3) CT performed on cycle ergometer can impair training-induced muscle hypertrophy even in individuals who were undergoing ST and/or ET for at least 1 year prior the beginning of the study.

Summary

In conclusion, the molecular basis of CT and the interference effect are incompletely depicted, which may be attributed to limitations/differences on previous investigations. Accordingly, all the disparities between study designs make the understanding of the molecular basis that control muscle hypertrophy in response to CT complex. In addition, it has been neglected that even for a high volume, frequency, and intensity CT regimen, it took 7–8 weeks to observe any interference effect [5]. Therefore, chronic studies aiming to scrutinize the effects of CT on molecular responses and the interference effect need to be longer in duration. Since we do not have convincing evidence supporting AMPK as a key factor regulating the interference effect, chronic studies should search for alternative pathways that may hamper myofibrillar protein synthesis when one undergoes CT. For example, the sirtuin family of NAD⁺-dependent deacetylases (SIRT) and Ca²⁺/calmodulin-dependent kinase II (CaMKII) can inhibit anabolic signaling pathways as well. In addition, advances in technology that allow researchers to investigate specific protein synthetic rates of contractile proteins (i.e., actin and myosin) are required to depict a better picture of the different anabolic training effects and to draw more specific conclusions. Those concerns suggest that further mechanistic studies are warranted to maximize the benefits of CT for special populations and to avoid the interference effect for athletic population.

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